Siyuan Mei

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Research Interests

Medical AI: reconstruction, synthesis, segmentation, and registration.
Computer vision: diffusion and flow-matching models, generative modeling, representation learning, and dense prediction.
LLMs & software engineering: LLM-assisted development, reproducible research tooling, and agentic systems.

Education

FAU Erlangen-Nürnberg — 2023–2027 (expected)
Ph.D. in Computer Science; Supervisor: Prof. Andreas Maier.

Peking University Health Science Center — May–Oct. 2026
Visiting Scholar, Artificial Intelligence for Medical Imaging (AIMI) Lab; Supervisor: Prof. Yixing Huang.

FAU Erlangen-Nürnberg — 2020–2023
M.Sc. in Medical Engineering; thesis grade: 1.0.

Zhejiang University of Science and Technology — 2016–2020
B.Eng. in Automation; Sino-German 2+3 Program; honored graduate scholarship.

Professional Experience

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) — 2023–2027
Ph.D. Student, Pattern Recognition Lab; Supervisor: Prof. Andreas Maier

- Fully funded through the ERC project *Advancing osteoporosis medicine by observing bone microstructure and remodelling using a 4D nanoscope*; developed **BigReg**, an efficient registration pipeline for high-resolution X-ray microscopy and light-sheet fluorescence microscopy.
- Conduct research in medical AI and computer vision, adapting foundation models and developing medical image generation methods with GANs, diffusion models, and flow matching.
- Teach *Flat-panel CT Reconstruction* and contribute lecture slides for *Vibe Coding: Software Engineering with LLMs*, a Springer book project.
- **Outcomes:** first-author work in Pattern Recognition and MICCAI; a Featured Paper in Journal of Medical Imaging; and an Oral Presentation at MICCAI 2026.

Peking University Health Science Center, AIMI Lab — May–Oct. 2026
Visiting Scholar; Supervisor: Prof. Yixing Huang
Artificial Intelligence for Medical Imaging Lab, Institute of Medical Technology, 240 Pathology Building, No. 38 Xueyuan Road, Haidian District, Beijing, China.

- Participated in the MICCAI BIC-MAC Challenge and trained a multimodal-to-CT generative model for PET attenuation correction; achieved first place on the validation-set CT metric.
- Built practical knowledge of clinical CT applications and imaging workflows; visited the ICT Center in Chongqing and attended MISC 2026 at Tsinghua University.

Selected Publications

- **[C.1]** **Siyuan Mei** et al. **Time Matters: Rethinking Diffusion and Flow Models in One-Step Medical Image Translation**. MICCAI 2026; **Early Accept**; **Oral Presentation**; first author.
- **[C.2]** Mengchen Fan et al., including Siyuan Mei. **Detail Consistent Stage-Wise Distillation for Efficient 3D MRI Segmentation**. MICCAI 2026; **Early Accept**; **Spotlight Presentation**.
- **[C.3]** Daiqi Liu et al., including Siyuan Mei. **Speech-Guided Multimodal Learning for Vocal Tract Segmentation in Real-Time MRI**. MICCAI 2026; **Oral Presentation**.
- **[C.4]** **Siyuan Mei** et al. **WING: A Window-Prior-Based Generative Network with Gated Inception for Cross-Modality CT Synthesis**. MICCAI MIART Workshop 2026; accepted workshop paper; first author.
- **[J.1]** **Siyuan Mei** et al. **Vision Transformer Hook for Dense Predictions**. Pattern Recognition 179 (2026), 113818; first author.

- **[J.2]** **Siyuan Mei** et al. **BigReg: An Efficient Registration Pipeline for High-Resolution X-Ray and Light-Sheet Fluorescence Microscopy**. Journal of Medical Imaging 12(5), 054004 (2025); **Featured Paper**; first author.
- **[C.5]** **Siyuan Mei**, F. Fan, A. Maier. **Metal Inpainting in CBCT Projections Using Score-based Generative Model**. IEEE ISBI 2023; first author.
- **[C.6]** **Siyuan Mei** (co-author). **TCCT: Trajectory-Conditioned CBCT Reconstruction for Sinusoidal Non-Circular Orbits with a Fourier Neural Operator**. MICCAI 2026; Poster.
- **[J.3]** Y. Sun et al., including Siyuan Mei. **Filter2Noise: A Framework for Interpretable and Zero-Shot Low-Dose CT Image Denoising**. Journal of Medical Imaging 13(2), 024004 (2026).
- **[J.4]** C. Ye et al., including Siyuan Mei. **Robustness and Stability Analysis of Differentiable Shift-Variant FBP for Cone-Beam CT under Challenging Acquisition Settings**. Machine Learning for Biomedical Imaging (2026).
- **[J.5]** M. Thies et al., including Siyuan Mei. **A Gradient-Based Approach to Fast and Accurate Head Motion Compensation in Cone-Beam CT**. IEEE Transactions on Medical Imaging (2024).
- **[J.6]** C. Ye et al., including Siyuan Mei. **DRACO: Differentiable Reconstruction for Arbitrary CBCT Orbits**. Physics in Medicine and Biology 70, 075005 (2025).
- **[C.7]** M. Gu et al., including Siyuan Mei. **Unsupervised Domain Adaptation Using Soft-Labeled Contrastive Learning with Reversed Monte Carlo Method for Cardiac Image Segmentation**. MICCAI 2024.
- **[C.8]** L. Pfaff et al., including Siyuan Mei. **No-New-Denoiser: A Critical Analysis of Diffusion Models for Medical Image Denoising**. MICCAI 2024.
- **[M.1]** Zirong Li, **Siyuan Mei** et al. **Generative Drifting for Conditional Medical Image Generation**. arXiv preprint (2026).
- **[M.2]** **Siyuan Mei**, Y. Xia, F. Fan. **GANeXt: A Fully ConvNeXt-Enhanced Generative Adversarial Network for MRI- and CBCT-to-CT Synthesis**. arXiv preprint (2025).

Honors & Awards

- **1st place, Task 2** and **4th place, Task 1**, MICCAI SynthRAD2025 Challenge, 2025.
- **Two merit-based scholarships** during M.Sc. studies at FAU Erlangen-Nürnberg.
- Fully funded Ph.D. through the ERC 4D Nanoscope project, FAU, 2023–2027.
- Honored graduate scholarship, Zhejiang University of Science and Technology, 2020.

Teaching & Academic Service

Flat-panel CT Reconstruction FAU
Summer semesters 2024–2026.

Vibe Coding: Software Engineering with LLMs FAU
Summer semester 2026; lecture slides accompanying a Springer book.

Journal service: Technical Editor, Journal of Medical Imaging; reviewer, Pattern Recognition and Journal of Machine Learning for Biomedical Imaging (MELBA).
Conference service: Reviewer, MICCAI.

Additional Information

Languages: Chinese (native), English (professional), German (C1 certificate).
Interests: fitness and tennis (level 2.5). **MBTI (self-reported):** ENTJ-A.
Career conversations: I am open to research conversations, collaborations, and job-related discussions in medical imaging, computer vision, and LLM-assisted software engineering; I appreciate hearing about suitable opportunities.
Open-source and teaching resources: WING, JiR, ViT-Hook, and Vibe Coding lecture slides accompanying a Springer book.